

TOBB University of Economics and Technology

Department of Electrical and Electronics Engineering

ELE 224L – Electronic Circuits Laboratory

Experiment 4 – Preliminary Work: BJT Circuits

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1 INTRODUCTION

1.1 Objective of the Experiment

The objective of this experiment is to investigate the current–voltage characteristics of the bipolar junction transistor (BJT) and to learn the DC biasing methods. The BC238A transistor is used to find the operating point of two biasing circuits, to see how the collector current depends on V_{CE} and on the base drive, and to compare two biasing schemes in terms of their sensitivity to the current gain β. The ability to diagnose wrong measurements of V_{CE} is also practised.

1.2 Benefits of the Experiment

The experiment shows how a BJT is analysed with the assumed-region method (assume active region, calculate, verify the assumption, switch to saturation if the assumption fails), why a bias circuit with an emitter resistor is preferred, and how measured voltages reveal wiring faults. It also trains the correct reading of a datasheet and the correct placement of a TO-92 transistor on a breadboard. These skills are directly used in the BJT amplifier experiment.

2 CALCULATIONS AND SIMULATIONS

2.1 BJT Properties

The parameters of the BC238A were taken from the datasheet of the BC237/238/239 family (Fairchild Semiconductor), and they are listed in Table 2.1.1. The current gain of the class A transistors is specified as $h_{FE}=120$–220 at $V_{CE}=5$ V and $I_C=2$ mA; the average value of 170 is used as β. The base–emitter turn-on voltage is typically 0.62 V (0.55–0.70 V) at the same bias. The collector–emitter saturation voltage is 0.2 V at most ($I_C=10$ mA, $I_B=0.5$ mA; typical 0.07 V), and 0.2 V is used as $V_{CE(sat)}$ in the calculations.

Table 2.1.1 BC238A parameters and TO-92 pin order from the datasheet

Parameter	Value	Condition
β (class A, average)	170 (range 120–220)	V_{CE} = 5 V, I_C = 2 mA
$V_{BE(on)}$ (typical)	0.62 V (min 0.55 V, max 0.70 V)	V_{CE} = 5 V, I_C = 2 mA
$V_{CE(sat)}$ (maximum)	0.2 V (typical 0.07 V)	I_C = 10 mA, I_B = 0.5 mA
Pin 1 / Pin 2 / Pin 3	Collector / Base / Emitter	TO-92, flat face towards the viewer, leads down

The pin order is C–B–E for the Fairchild datasheet. Some other manufacturers use E–B–C for the same package, so the pin order of the physical part should be checked before the circuit is powered.

2.2 BJT Circuit 1 on the Breadboard

BJT circuit 1 (Figure 2.2.1) was drawn on the breadboard as shown in Figure 2.2.2. The transistor is placed with its three legs on three adjacent columns (collector on column 10, base on column 11, emitter on column 12). R_C is connected between the collector column and a column that is wired to the $+V_{CC}$ rail, R_E is connected between the emitter column and a column that is wired to the ground rail, and R_B is connected between the base column and the V_{BB} supply lead. The connections of the drawing were verified automatically against the circuit diagram.

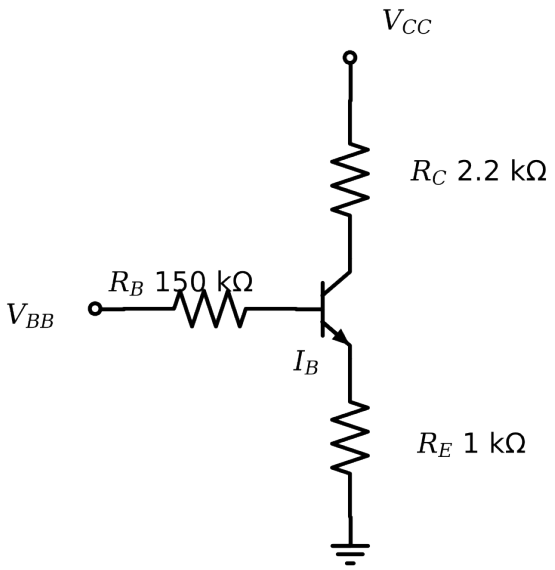


Figure 2.2.1 BJT circuit 1 of the experiment (R_B = 150 kΩ, R_C = 2.2 kΩ, R_E = 1 kΩ).

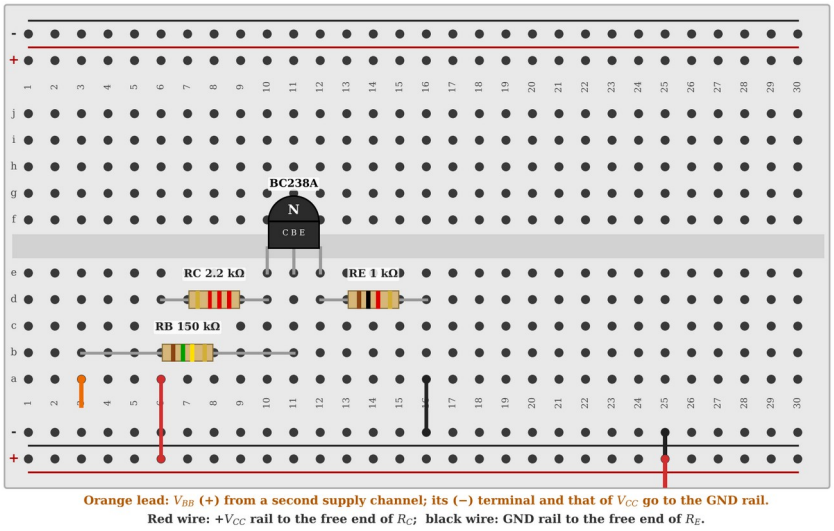


Figure 2.2.2 BJT circuit 1 drawn on the breadboard (V_{BB}, V_{CC} and GND points are indicated).

2.3 Q-Point Calculations of BJT Circuit 1

The operating point of circuit 1 was calculated for $V_{BB}=1$, 2 and 5 V and for the V_{CC} values of the experiment sheet with the model $\beta=170$, $V_{BE(on)}=0.62$ V, $V_{CE(sat)}=0.2$ V and no Early effect. The base-loop equation (2.1) gives the base current when the transistor is in the active region, and the remaining quantities follow from (2.2) and (2.3).

$$V_{BB}=I_B R_B+V_{BE(on)}+I_E R_E,\ I_E=(\beta+1)I_B \Rightarrow I_B=\frac{V_{BB}-V_{BE(on)}}{R_B+(\beta+1)R_E} \tag{2.1}$$

$$I_C=\beta I_B,\ I_E=(\beta+1)I_B \tag{2.2}$$

$$V_{CE}=V_{CC}-I_C R_C-I_E R_E \tag{2.3}$$

The active-region assumption is verified by $V_{CE}\geq V_{CE(sat)}$. If it fails, the transistor is in saturation: $V_{CE}=V_{CE(sat)}$, and I_B and I_C are found from the two loop equations given in (2.4); $I_C=\beta I_B$ no longer holds. If the supply is so small that even this solution gives $I_C\leq 0$, the collector current is practically zero and $V_{CE}=V_{CC}$.

$$\begin{aligned}(R_B+R_E)I_B+R_E I_C &\quad ¿\, V_{BB}-V_{BE(on)}\\ R_E I_B+(R_C+R_E)I_C &\quad ¿\, V_{CC}-V_{CE(sat)}\end{aligned} \tag{2.4}$$

Sample calculation 1 (active region). For $V_{BB}=2$ V and $V_{CC}=10$ V: $I_B=(2-0.62)/(150\,\text{k}\Omega+171\cdot 1\,\text{k}\Omega)=4.299$ μA, $I_C=170\cdot 4.299\,\mu\text{A}=0.7308$ mA, $I_E=0.7351$ mA, and $V_{CE}=10-0.7308\cdot 2.2-0.7351\cdot 1=7.657$ V >0.2 V, so the assumption is valid.

Sample calculation 2 (saturation). For $V_{BB}=5$ V and $V_{CC}=4$ V: the active-region result would be $V_{CE}=4-7.44<0.2$ V, so the transistor is in saturation, and (2.4) gives $I_C=1.181$ mA, $V_{CE}=0.2$ V. The other cells are calculated in the same way.

Table 2.3.1 Table 1 values for V_{BB} = 1 V (calculation and SPICE)

V_{CC} (V)	0	0.2	0.5	1	2	4	6	8	10	15	20
V_{CE} calc. (V)	0	0.2	0.2	0.355	1.35	3.35	5.35	7.35	9.35	14.4	19.4
I_C calc. (mA)	0	0	0.093	0.201	0.201	0.201	0.201	0.201	0.201	0.201	0.201
V_{CE} SPICE (V)	0.0052	0.073	0.114	0.262	1.26	3.26	5.26	7.26	9.26	14.3	19.3
I_C SPICE (mA)	−0.0028	0.039	0.12	0.23	0.231	0.231	0.231	0.231	0.231	0.231	0.231

Table 2.3.2 Table 1 values for V_{BB} = 2 V (calculation and SPICE)

V_{CC} (V)	0	0.2	0.5	1	2	4	6	8	10	15	20
V_{CE} calc. (V)	0	0.2	0.2	0.2	0.2	1.66	3.66	5.66	7.66	12.7	17.7
I_C calc. (mA)	0	0	0.091	0.248	0.561	0.731	0.731	0.731	0.731	0.731	0.731
V_{CE} SPICE (V)	0.0088	0.05	0.076	0.101	0.146	1.61	3.61	5.61	7.61	12.6	17.6
I_C SPICE (mA)	−0.0059	0.044	0.13	0.279	0.578	0.744	0.744	0.744	0.744	0.744	0.744

Table 2.3.3 Table 1 values for V_{BB} = 5 V (calculation and SPICE)

V_{CC} (V)	0	0.2	0.5	1	2	4	6	8	10	15	20
V_{CE} calc. (V)	0	0.2	0.2	0.2	0.2	0.2	0.2	0.564	2.56	7.56	12.6
I_C calc. (mA)	0	0	0.085	0.241	0.555	1.18	1.81	2.32	2.32	2.32	2.32
V_{CE} SPICE (V)	0.012	0.033	0.05	0.067	0.088	0.117	0.149	0.571	2.57	7.57	12.6
I_C SPICE (mA)	−0.013	0.043	0.132	0.283	0.59	1.21	1.82	2.32	2.32	2.32	2.32

The results are plotted in Figure 2.3.1. In the active region (large V_{CC}), I_C is constant for a given V_{BB} (0.201, 0.731 and 2.32 mA for V_{BB} = 1, 2, 5 V) and V_{CE} increases linearly with V_{CC}, since the model has no Early effect; the three horizontal lines are the output characteristics of the transistor for three different base currents ($I_B=1.18$, 4.30 and 13.6 μA). When V_{CC} is decreased, the transistor enters the saturation region (the cells $V_{CC}\le 6$ V for $V_{BB}=5$ V, ≤ 2 V for $V_{BB}=2$ V and ≤ 0.5 V for $V_{BB}=1$ V), where V_{CE} is clamped at about 0.2 V and I_C falls linearly with V_{CC}; this is the steep part of the characteristic. For $V_{CC}=0$, all currents are zero.

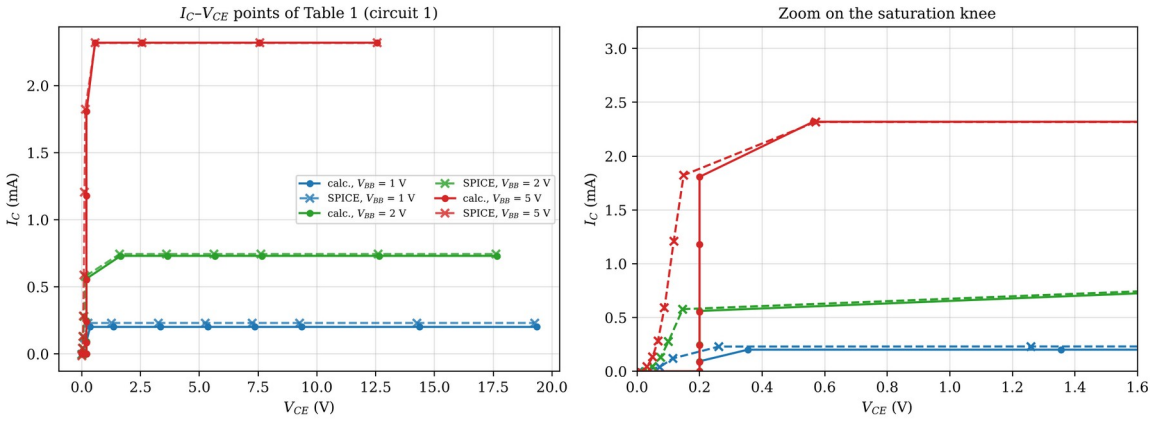


Figure 2.3.1 I_C–V_{CE} points of Table 1 (calculation: circles; SPICE: crosses); right: zoom on the saturation region.

The SPICE results (BC238A model with I_S fitted to $V_{BE}=0.62$ V at 2 mA, $\beta=170$, no Early effect; the model line is in Appendix A) are also given in the tables. In the active region, the largest difference in V_{CE} is 0.0948 V (V_{BB} = 1 V), 0.0432 V (2 V) and 0.00739 V (5 V), and the mean difference in I_C is +14.6%, +1.8% and -0.1%, respectively. The difference is largest for $V_{BB}=1$ V because the real V_{BE} depends logarithmically on the current: at $I_C\approx0.2$ mA the real V_{BE} is about 0.56 V, not 0.62 V, so the base current (which is proportional to $V_{BB}-V_{BE}$) is 15% larger than the calculated value. For $V_{BB}=5$ V, the term $V_{BB}-V_{BE}$ is large and the effect of the error in V_{BE} disappears. In the saturation region, SPICE gives V_{CE} between 0.03 and 0.15 V (it decreases with the base drive), whereas the model uses a constant 0.2 V; hence the saturation cells have a difference of up to 0.15 V in V_{CE} and a few percent in I_C.

2.4 Biasing Circuit 1 (No Emitter Resistor)

The circuit in Figure 2.4.1 has no emitter resistor. For $V_{BB}=5$ V and $V_{CC}=12$ V, the base current is $I_{BQ}=(V_{BB}-V_{BE})/R_B=(5-0.62)/150\,\mathrm{k}\Omega=29.20$ µA, the collector current is $I_{CQ}=\beta I_{BQ}=4.964$ mA, and the collector–emitter voltage is $V_{CEQ}=V_{CC}-I_{CQ}R_C=12-4.964\times2.2=1.079$ V, with $V_{BEQ}=0.62$ V. Since $V_{CEQ}>0.2$ V (and $V_{CEQ}>V_{BE}$, i.e., the collector junction is reverse biased), the active-region assumption is valid. The SPICE simulation gives $V_{BEQ}=0.6442$ V, $I_{BQ}=29.04$ µA, $I_{CQ}=4.937$ mA and $V_{CEQ}=1.140$ V; the differences from the calculation result from the dependence of V_{BE} on the current ($V_{BE}=0.644$ V at 5 mA).

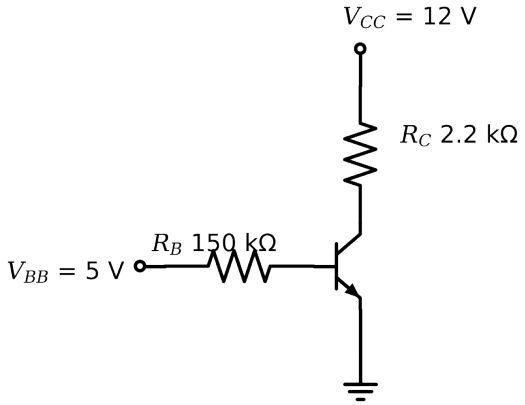


Figure 2.4.1 BJT biasing circuit 1 (no emitter resistor).

The operating point of this circuit is very sensitive to β, because $I_C = \beta I_B$ and I_B is fixed by V_{BB} and R_B. With the datasheet range of β for the class A transistors, I_C changes from 3.50 mA ($\beta = 120$) to 5.36 (saturation limit) mA ($\beta = 220$) in the active-region equations; for $\beta = 220$, the active-region result would give $V_{CE} < 0$, so the transistor actually saturates with $V_{CE} = 0.2$ V and $I_C = 5.36$ mA, and the amplifier would be unusable. This is shown in the left graph of Figure 2.5.2.

2.5 Biasing Circuit 2 (With Emitter Resistor)

The circuit in Figure 2.5.1 has $R_B = 22$ kΩ, $R_C = 1.2$ kΩ and $R_E = 1$ kΩ. The calculation with (2.1)–(2.3) for $V_{BB} = 5$ V and $V_{CC} = 12$ V gives $I_{BQ} = 4.38 / (22\,\text{k}\Omega + 171\,\text{k}\Omega) = 22.69$ μA, $I_{CQ} = 3.858$ mA, $I_{EQ} = 3.881$ mA, $V_{BEQ} = 0.62$ V, and $V_{CEQ} = 12 - 3.858 \times 1.2 - 3.881 \times 1 = 3.490$ V. The transistor is in the active region. The SPICE simulation gives $V_{BEQ} = 0.6376$ V, $I_{BQ} = 22.60$ μA, $I_{CQ} = 3.843$ mA and $V_{CEQ} = 3.524$ V.

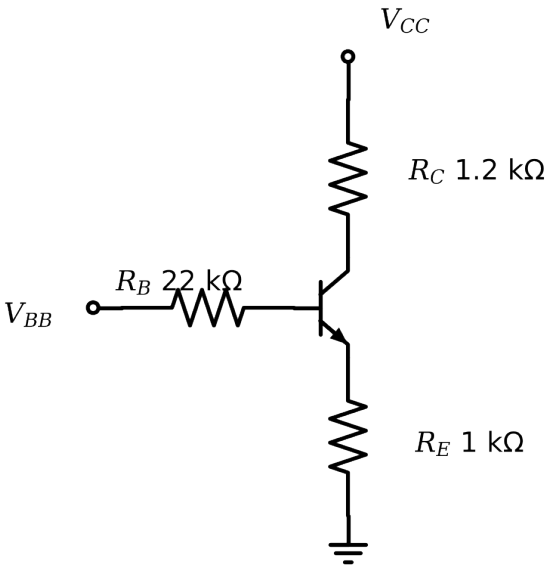


Figure 2.5.1 BJT biasing circuit 2 (with emitter resistor).

Table 2.5.1 Sensitivity of the two bias circuits to β (V_{BB} = 5 V, V_{CC} = 12 V)

β	**Circuit 1: I_C (mA)**	**Circuit 1: V_{CE} (V)**	**Circuit 2: I_C (mA)**	**Circuit 2: V_{CE} (V)**
120	3.50	4.29	3.68	3.88
170	4.96	1.08	3.86	3.49
220	5.36	0.200 (saturated)	3.97	3.26

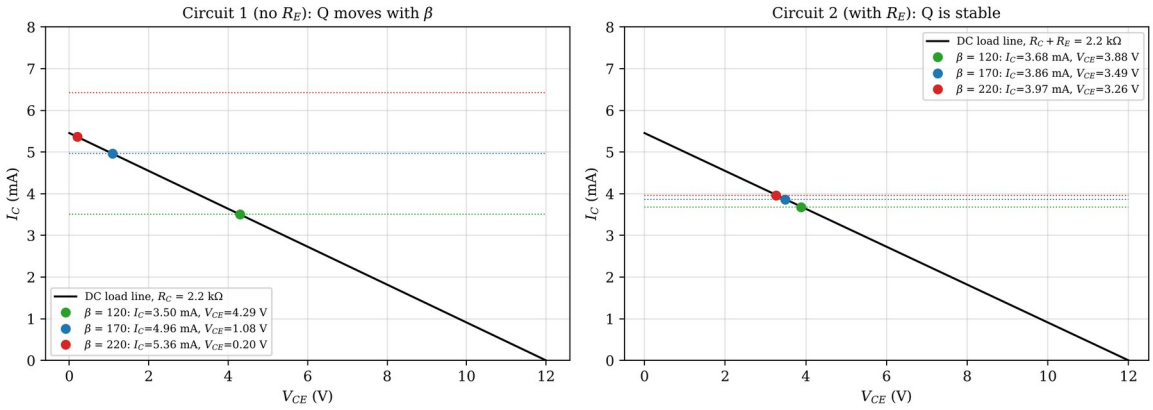


Figure 2.5.2 DC load lines and operating points of the two bias circuits for β = 120, 170 and 220 (calculation).

In circuit 2, the emitter resistor provides negative feedback: if I_C tries to increase, the voltage drop on R_E increases, the base–emitter voltage decreases, and I_B decreases. As Table 2.5.1 shows, a change of β from 120 to 220 changes I_C only from 3.68 to 3.97 mA (8%), whereas I_C of circuit 1 changes by 37% and finally saturates. The operating point is stable because $(\beta+1)R_E = 171$ kΩ is large compared with R_B, so that $I_C \approx (V_{BB} - V_{BE})/R_E$ is almost independent of β.

2.6 Possible Reasons for Wrong V_{CEQ} Readings

The reasons are listed item by item for biasing circuit 2 of Figure 2.5.1 (the correct value is $V_{CEQ} \approx 3.5$ V). The values in Table 2.6.1 were obtained from SPICE simulations of each fault.

Table 2.6.1 SPICE results of the fault conditions in BJT biasing circuit 2 (V_{BB} = 5 V, V_{CC} = 12 V)

Condition	V_{CE} **(V)**	V_{BE} **(V)**	I_C **(mA)**
Normal operation (reference)	3.52	0.638	3.84
V_{BB} = 0 V (base supply missing)	12.0	0	0
R_B open (or base lead not inserted)	12.0	≈ 0 (floating)	0
Collector and emitter pins exchanged	11.4	0.564	0.185
R_E open (emitter floating)	7.11	0.107	0
R_C open (collector floating)	0.0179	0.583	0
V_{CC} = 0 V (collector supply missing)	0.0107	0.576	−0.0941
R_E shorted by a wire	0.121	0.668	9.90
R_C = 12 kΩ instead of 1.2 kΩ	0.0556	0.611	0.907
Collector–emitter short circuit	≈ 0	−0.455	5.45

If V_{CEQ} is measured as 12 V, no current flows through R_C and R_E ($I_C \approx 0$), i.e., the transistor is in cut-off or the collector current path is broken. The possible reasons and the corresponding actions are:

- No base current: V_{BB} is not applied or is 0 V (supply channel off, current limit at zero, ground wire not common). Action: measure V_B with respect to ground; if it is 0 V, check the V_{BB} channel and its wiring.
- Open R_B or an unconnected base: a loose resistor leg, a hole in the wrong column, or a broken wire. Action: measure the voltage across R_B (it should be about 0.5 V, i.e., $I_B R_B = 22.7\,\mu\text{A} \times 22\,\text{k}\Omega$); if $V_B = V_{BB}$ and $V_{BE} \approx 0$, the base is connected but the emitter path is open; if the voltage at the base side is 0 V, the base is not connected.
- Transistor inserted incorrectly or damaged (open base–emitter junction): a wrong pin order places the base on a wrong column. Action: check the pin order of the part (C–B–E), measure V_{BE} (it should be 0.6–0.7 V when the transistor conducts), and replace the transistor if the junction does not conduct.
- Exchanged collector and emitter pins: the transistor works in the inverse mode with a very small current gain, and V_{CE} is close to 12 V (11.4 V in the simulation, with $V_{BE} \approx 0.56$ V and a small I_C). Action: exchange the two pins.

- A measurement error: the voltmeter is connected between V_{CC} and a point instead of between collector and emitter, or R_E is open (then V_{CE} is not 12 V but 7.1 V in the simulation, and $V_{BE} \approx 0.1$ V, which distinguishes this case).

If V_{CEQ} is measured as 0 V, the transistor is in deep saturation or there is a short circuit between the collector and the emitter. The possible reasons and the corresponding actions are:

- Excessive base drive or too large a collector resistor: R_E shorted by a wire (0.121 V in the simulation, $I_C = 9.9$ mA), or a wrong R_C value such as 12 kΩ (0.0556 V). Action: measure each resistor with an ohmmeter and compare with the colour code; remove the unintended wire.
- A short circuit between collector and emitter (a wire or a solder bridge on the breadboard, or a damaged transistor). In this case, V_{BE} is typically negative or abnormal (−0.45 V in the simulation) and I_C is about $V_{CC}/(R_C + R_E) = 5.5$ mA. Action: remove the short, or replace the transistor if it is damaged.
- V_{CC} is not applied (0 V): the supply channel is off or its wire is disconnected (0.0107 V in the simulation, with a small reverse current in R_C). Action: measure V_{CC} at the top of R_C.
- R_C is open or not connected to the collector: the collector floats and the forward-biased collector junction pulls it to the emitter voltage (0.0179 V in the simulation, $I_C = 0$, $V_{BE} \approx 0.58$ V). Action: check the continuity of R_C and the column of the collector.
- A measurement error, such as both probes on the same node or the ground lead of the voltmeter on a floating ground. Action: repeat the measurement with the probes directly on the collector and emitter pins.

In every case, the systematic approach is to measure V_B, V_E and V_C with respect to ground and to compare them with the expected values ($V_B \approx 4.50$ V, $V_E \approx 3.88$ V, $V_C \approx 7.37$ V) in order to find the first node where the measurement deviates.

3 CONCLUSIONS

The calculated and the simulated results of Table 1 agree well in the active region when the base drive is large ($V_{BB} = 5$ V: the difference in I_C is -0.1% and in V_{CE} 0.00739 V), and the difference grows when the base drive is small ($V_{BB} = 1$ V: +14.6% in I_C), because the constant $V_{BE(on)} = 0.62$ V of the model is valid only near 2 mA. In the saturation region, the constant

$V_{CE(sat)} = 0.2$ V of the model is larger than the SPICE values (0.03–0.15 V), which is a limitation of the simple model rather than a calculation error. In the bias circuits, the circuit without R_E has an operating point that depends strongly on β (and even saturates for $\beta \geq 184$, which is inside the datasheet range of 120–220), whereas the circuit with R_E changes its collector current by only about 8% over the same range, which is why emitter feedback is used in practical amplifiers. The fault analysis showed that $V_{CE} = 12$ V and $V_{CE} = 0$ V can each arise from several different faults, and that the additional measurements of V_{BE} and the node voltages are necessary to separate them.

4 REFERENCES

1. Fairchild Semiconductor. (2002). BC237/238/239 NPN epitaxial silicon transistor [Data sheet, Rev. A2]. Retrieved October 11, 2026, from https://wmsc.lcsc.com/wmsc/upload/file/pdf/v2/lcsc/2309151357_onsemi-BC237_C3201628.pdf
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5 APPENDICES

5.1 Appendix A: SPICE Model and Netlists

The BC238A model was fitted to the datasheet typical values (I_S for $V_{BE} = 0.62$ V at 2 mA, $\beta_F = 170$); $\beta_R = 1$, $R_B = 30\,\Omega$ and $R_C = 1\,\Omega$ approximately reproduce the saturation voltages of the datasheet, and the Early voltage is infinite as required by the experiment sheet. The library model of the BC238 in another simulation program may differ slightly.

Code A.1 BJT model line used in the ngspice simulations

```
.model BC238A NPN(IS=7.7754e-14 BF=170 BR=1 RB=30 RC=1 CJE=8p CJC=3.5p TF=0.5n)
```

Netlist A.2 BJT circuit 1 (V_{BB} = 5 V, V_{CC} = 10 V)

```
Vbb vbb 0 DC 5
Vcc vcc 0 DC 10
RB vbb b 150000
RC vcc c 2200
RE e 0 1000
Q1 c b e BC238A
.model BC238A NPN(IS=7.7754e-14 BF=170 BR=1 RB=30 RC=1 CJE=8p CJC=3.5p TF=0.5n)
```

```
.op
.end
```

Netlist A.3 BJT biasing circuit 2

```
Vbb vbb 0 DC 5
Vcc vcc 0 DC 12
RB vbb b 22k
RC vcc c 1.2k
RE e 0 1k
Q1 c b e BC238A
.model BC238A NPN(IS=7.7754e-14 BF=170 BR=1 RB=30 RC=1 CJE=8p CJC=3.5p TF=0.5n)

.op
.end
```